

MATH255: Honours Analysis 2

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Abstract

1 Metric Spaces

1.1 Continuous functions on compact sets

2 Infinite Series

Definition 1 (Partial sum). Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. $k_0 \in \mathbb{Z}$ and $(a_k)_{k \geq k_0}$ by a sequence in X . For every $n \geq k_0$, $S_n = \sum_{k=k_0}^n a_k$ is called the partial sum of order n of $(a_k)_k$.

If $\lim_{n \rightarrow \infty} S_n = S$ in $(X, \|\cdot\|)$, for some $S \in X$, then we write

$$\sum_{k=k_0}^{\infty} a_k = S \quad (1)$$

and we say that the series $\sum_{k=k_0}^{\infty} a_k$ and that S is its sum. If $(S_n)_{n \geq k_0}$ does not converge, then we say that the series $\sum_{k=k_0}^{\infty} a_k$ diverges.

2.1 Series with nonnegative term

Theorem 1 (Direct Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $0 \leq a_k \leq b_k$ for all $k \geq k_0$. Then

1. If $\sum_{k=k_0}^{\infty} b_k$ converges, then $\sum_{k=k_0}^{\infty} a_k$ converges and $\sum_{k=k_0}^{\infty} a_k \leq \sum_{k=k_0}^{\infty} b_k$.
2. If $\sum_{k=k_0}^{\infty} a_k$ diverges, then $\sum_{k=k_0}^{\infty} b_k$ diverges.

Corollary 1 (Limit Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ and $b_k > 0$ for all $k \geq k_0$, and $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \ell \in (0, \infty)$. Then $\sum_{k=k_0}^{\infty} a_k$ converges if and only if $\sum_{k=k_0}^{\infty} b_k$ converges.

Theorem 2 (Root Test).

Proposition 1. Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ for all $k \geq k_0$.

1. If $\limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$, then $\sum_{k=k_0}^{\infty} a_k$ converges.
2. If $\liminf_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$, then $\sum_{k=k_0}^{\infty} a_k$ diverges.

Proof: (1) Assume that $R = \limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$. Then $\forall \varepsilon \in (0, 1 - R)$, $\exists n_\varepsilon \in \mathbb{N}$ such that $\forall k \geq n_\varepsilon$, $\sup_{k \geq n} \frac{a_{k+1}}{a_k} < R + \varepsilon < 1$, which gives for

all $k \geq n_\varepsilon$, $a_{k+1} < (R + \varepsilon)a_k < (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$. When $\varepsilon \in (0, 1 - R)$, $0 < R + \varepsilon < 1$, so $\sum_{k=n_\varepsilon}^{\infty} (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$ converges, which by direct comparison implies that $\sum_{k=k_0}^{\infty} a_k$ converges.

(2) Assume that $R = \liminf \frac{a_{k+1}}{a_k} > 1$. Then $\exists k_1 \geq k_0$ such that $\forall k \geq k_1$, $\inf_{k \geq n} \frac{a_{k+1}}{a_k} > R - \varepsilon > 1$, i.e. $a_{k+1} > a_k$, which implies $a_k \not\rightarrow 0$ as $k \rightarrow \infty$, thus $\sum_{k=k_0}^{\infty} a_k$ diverges. $\square$

Now we focus on series in $(\mathbb{R}, |\cdot|)$ with terms of different sign.

Theorem 3 (Alternating Series Test). *Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, |\cdot|)$ such that $0 \leq a_{k+1} \leq a_k$ for all $k \geq k_0$ and $\lim_{k \rightarrow \infty} a_k = 0$. Then $\sum_{k=k_0}^{\infty} a_k$ converges and*

$$0 \leq \sum_{k=k_0}^{\infty} (-1)^{k-k_0} a_k \leq a_{k_0} \quad (2)$$

Example 1

$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$ converges since $0 \leq \frac{1}{k+1} \leq \frac{1}{k}$ for all $k \geq 1$ and $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$.

Proof: By observing that

$$\sum_{k=k_0}^{\infty} (-1)^k a_k = \sum_{k=0}^{\infty} (-1)^{k+k_0} a_{k+k_0} \quad (3)$$

$$= (-1)^{k_0} \sum_{k=0}^{\infty} (-1)^k a_{k+k_0} \quad (4)$$

we can assume that $k_0 = 0$.

For every $n \in \mathbb{N}$,

$$S_{2n} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} \quad (5)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + a_{2n} \quad (6)$$

$$\geq 0 \quad (7)$$

$$S_{2n+1} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} - a_{2n+1} \quad (8)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + (a_{2n} - a_{2n+1}) \quad (9)$$

$$\leq 0 \quad (10)$$

Definition 2 (Cauchy product). *Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$.*

For each $n \in \mathbb{N}$, define

$$c_n = \sum_{k=0}^n a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 \quad (11)$$

Then the series $\sum_{n=0}^{\infty} c_n$ is called the Cauchy product of $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$.

Proposition 2. If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ are series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$ such that $\sum_{k=0}^{\infty} a_k$ converges and $\sum_{k=0}^{\infty} b_k$ converges (not necessarily absolutely). Then the Cauchy product $\sum_{n=0}^{\infty} c_n$ converges and $\sum_{n=0}^{\infty} c_n = (\sum_{k=0}^{\infty} a_k)(\sum_{k=0}^{\infty} b_k)$.

Example 2 (Counter example of Cauchy product of two conditionally converging series)

Consider $a_k = b_k = \frac{(-1)^k}{\sqrt{k+1}}$. Then $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ converge by the alternating series test and p -series convergence.

In this case:

$$|c_n| = \left| \sum_{k=0}^n a_k b_{n-k} \right| \quad (12)$$

$$(13)$$

Definition 3 (Power series). Let $(a_k)_k$ be a sequence in $\mathbb{R}$ or $\mathbb{C}$ and $x_0 \in \mathbb{R}$ or $\mathbb{C}$. The function $x \mapsto \sum_{k=0}^{\infty} a_k (x - x_0)^k$ is defined at points where it converges, is called a power series.

Proposition 3 (Radius of convergence). Let $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ be a power series in $\mathbb{R}$ or $\mathbb{C}$. Define $R = \frac{1}{\limsup_{k \rightarrow \infty} \sqrt[k]{|a_k|}}$ with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$. Then R is called the radius of convergence of the power series, and

$$\text{if } |x - x_0| < R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ converges,} \quad (14)$$

$$\text{if } |x - x_0| > R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ diverges.} \quad (15)$$

Example 3

For every $x \in (0, 2)$, we can write $\ln x = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x - 1)^k$.

Indeed, let $\tilde{\ln} x = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} (x - 1)^k$. We want to show that $\tilde{\ln} x = \ln x$, i.e., $\tilde{\ln} e^y = y$ where $y = \ln x$. Observe that

$$\tilde{\ln}'(x) = \sum_{k=1}^{\infty} (-1)^{k-1} (x - 1)^{k-1} = \sum_{k=1}^{\infty} (1 - x)^{k-1} \quad (16)$$

$$= \sum_{k=0}^{\infty} (1 - x)^k \quad (17)$$

$$= \frac{1}{1 - (1 - x)} = \frac{1}{x} \quad (18)$$

More, this is true for all $x \in (0, 2)$ since the radius of convergence of $\sum_{k=0}^{\infty} (1 - x)^k$ is 1. It follows that

$$\frac{d}{dy} [\tilde{\ln} e^y - y] = e^y \tilde{\ln}' e^y - 1 = e^y \frac{1}{e^y} - 1 = 0 \quad (19)$$

hence $\tilde{\ln} e^y - y = \tilde{\ln} e^0 - 0 = 0$, i.e. $\tilde{\ln} x = \ln x$. $\square$

Definition 4 (Taylor series). Let I be an interval in $\mathbb{R}$, $x_0 \in I$ and $f : I \rightarrow \mathbb{R}$ be infinitely differentiable at x_0 . Then the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (20)$$

is called the Taylor series of f at x_0 .

Theorem 4 (Taylor's theorem with mean value reminder).

Let $x, x_0 \in \mathbb{R}$, $x \neq x_0$, $I = \begin{cases} [x_0, x] & \text{if } x > x_0 \\ [x, x_0] & \text{if } x < x_0 \end{cases}$, $\text{int}(I) = \begin{cases} (x_0, x) & \text{if } x > x_0 \\ (x, x_0) & \text{if } x < x_0 \end{cases}$ and $f : I \rightarrow \mathbb{R}$ be n times continuously differentiable in I (i.e. $f^{(n)}$ exists and is continuous in I). and $n + 1$ times differentiable in $\text{int}(I)$.

Then, $\exists x' \in \text{int}(I)$ such that

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{\text{partial sum of Taylor series at } x_0} + \underbrace{\frac{f^{(n+1)}(x')}{(n+1)!} (x - x_0)^{n+1}}_{\text{mean value reminder}} \quad (21)$$

Define $g(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (t - t)^k$. We want to show that $g(x_0) = f(x) - \frac{1}{(n+1)!} f^{(n+1)}(x') (x - x_0)^{n+1}$ for some $x' \in \text{int}(I)$.

In particular, $g(x) = f(x) - f(x) = 0$.

g is continuous on I and differentiable on $\text{int}(I)$ (by assumption on f) and

$$g'(t) = \sum_{k=0}^n \frac{f^{(k+1)}(t)}{k!} (t - t)^k \quad (22)$$

$$= \frac{f^{(n+1)}(t)}{n!} (t - t)^n \quad (23)$$

$$= \frac{f^{(n+1)}(t)}{(n+1)!} (t - x_0)^{n+1} - \frac{f^{(n+1)}(t)}{(n+1)!} (t - x)^{n+1} \quad (24)$$

Now define

$$h(t) = g(t)(x - x_0)^{n+1} - g(x_0)(x - t)^{n+1} \quad (25)$$

so that

$$h(x) = 0 \quad (26)$$

$$h(x_0) = 0 \quad (27)$$

Since h is continuous on I and differentiable on $\text{int}(I)$ (by assumption

We've proven that every power series $f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$ is equal to its Taylor series at $x_0 \in (x_0 - R, x_0 + R)$.

However, this is not true in general where f is only assumed to be infinitely differentiable at x_0 . For example,

Example 4

The function $x \mapsto e^{-\frac{1}{x^2}}$ for $x \neq 0$ and 0 for $x = 0$ is infinitely differentiable in $\mathbb{R}$ and such that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$. That means that its Taylor series at 0 is the zero series.

Thus the function itself is not equal to its Taylor series at 0.

The point here is to construct a function h such that it vanishes at both x and x_0 so that we can apply Rolle's theorem to it.

tion on g), and

$$h'(t) = g'(t)(x - x_0)^{n+1} + (n+1)g(x_0)(x - t)^n \quad (28)$$

$$(29)$$

By Rolle's Theorem, it follows that

$$\exists x' \in \text{int}(I) \text{ such that } h'(x') = 0 \quad (30)$$

3 Riemann-Stieltjes integral

Proposition 4. $f \in \mathcal{R}([a, b], \alpha)$ iff $\forall \varepsilon > 0, \exists P_\varepsilon$ (partition of $[a, b]$) such that $U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) < \varepsilon$.

Proof: ($\Rightarrow$) Assume that $f \in \mathcal{R}([a, b], \alpha)$. Then by properties of *inf* and *sup*, $\forall \varepsilon > 0, \exists P_\varepsilon, P'_\varepsilon$ partition of $[a, b]$ such that

$$\int_a^b f(x) d\alpha - \varepsilon < L(P_\varepsilon, f, \alpha) = \int_a^b f(x) d\alpha \leq \overline{\int_a^b f(x) d\alpha} \leq U(P_\varepsilon, f, \alpha) < \overline{\int_a^b f(x) d\alpha} + \varepsilon \quad (31)$$

Let $P_\varepsilon^* = P_\varepsilon \cup P'_\varepsilon$. Then since $P_\varepsilon^* \supset P_\varepsilon$ and $P_\varepsilon^* \supset P'_\varepsilon$,

$$U(P_\varepsilon^*, f, \alpha) - L(P_\varepsilon^*, f, \alpha) \leq U(P'_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) \quad (32)$$

$$< (\overline{\int_a^b f(x) d\alpha} + \frac{1}{2}\varepsilon) - (\int_a^b f(x) d\alpha - \frac{1}{2}\varepsilon) \quad (33)$$

$$= \overline{\int_a^b f(x) d\alpha} - \int_a^b f(x) d\alpha + \varepsilon \quad (34)$$

$$= \varepsilon \quad (35)$$

where $\overline{\int_a^b f(x) d\alpha} - \int_a^b f(x) d\alpha = 0$ since $f \in \mathcal{R}([a, b], \alpha)$.

($\Leftarrow$) Assume that $\forall \varepsilon > 0, \exists P_\varepsilon$ partition of $[a, b]$ such that $U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) < \varepsilon$. We have

$$L(P_\varepsilon, f, \alpha) \leq \int_a^b f(x) d\alpha \leq \overline{\int_a^b f(x) d\alpha} \leq U(P_\varepsilon, f, \alpha) \quad (36)$$

It follows that

$$0 \leq \overline{\int_a^b f(x) d\alpha} - \int_a^b f(x) d\alpha \leq U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) \leq \varepsilon \quad (37)$$

Letting $\varepsilon \rightarrow 0$, we have

$$\overline{\int_a^b f(x) d\alpha} - \int_a^b f(x) d\alpha = 0 \quad (38)$$

therefore $f \in \mathcal{R}([a, b], \alpha)$. □

Look at Question 2 in Assignment 6.

Proposition 5. Let $f, \alpha: [a, b] \rightarrow \mathbb{R}$ be such that f is bounded in $[a, b]$ and α increasing on $[a, b]$.

Let D be the set of points in $[a, b]$ where f is not continuous. Assume that D is countable and that α is continuous at every point in D . Then $f \in \mathcal{R}([a, b], \alpha)$.

Let $\varepsilon > 0$. For every $x \in [a, b] \setminus D$, since f is continuous at x , there exists $\delta_x > 0$ such that $|f(y) - f(x)| < c\varepsilon$ for all $y \in (x - \delta_x, x + \delta_x) \cap [a, b]$, where the constant $c > 0$ will be determined later.

Since D is countable, $\exists (x_k)_k$ in $[a, b]$ such that $D = \{x_k : k \in \mathbb{N}\}$. Since α is continuous at x_k , there exists $\delta_k > 0$ such that $|\alpha(y) - \alpha(x_k)| < c_k \varepsilon$ for all $y \in (x_k - \delta_k, x_k + \delta_k) \cap [a, b]$, where the constant $c_k > 0$ will be determined later.

Since $[a, b]$ is compact, $[a, b] \subset \bigcup_{x \in [a, b]} (x - \delta_x, x + \delta_x)$, there exists a finite subcover, i.e. $\exists y_1 < \dots < y_{N_\varepsilon} \in [a, b]$ such that

$$[a, b] \subset \bigcup_{i=1}^{N_\varepsilon} (y_i - \delta_{y_i}, y_i + \delta_{y_i}) \quad (39)$$

δ_x is defined for every $x \in [a, b]$, since either it's defined by the continuity of f at x (if $x \in [a, b] \setminus D$) or by the continuity of α at x (if $x \in D$).

By ordering the points $a, b, y_i - \delta_{y_i}$ (if ≥ 0), and $y_i + \delta_{y_i}$ (if $\leq b$), we then obtain a partition $P_\varepsilon = \{x_0, \dots, x_{m_\varepsilon}\}$ such that $\forall j \in \{0, \dots, m_\varepsilon\}$, $\exists i \in \{1, \dots, N_\varepsilon\}$ such that $(x_{j-1}, x_j) \subset (y_i - \delta_{y_i}, y_i + \delta_{y_i}) \cap [a, b]$.

We again separate this into two cases (depending on whether $y_i \in D$ or not):

- If $y_i \notin D$, then

$$M_j - m_j = \sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \quad (40)$$

$$\leq \left(\sup_{x \in [x_{j-1}, x_j]} f(x) - f(y_i) \right) + \left(f(y_i) - \inf_{x \in [x_{j-1}, x_j]} f(x) \right) \quad (41)$$

$$< 2c\varepsilon \quad (42)$$

- If $y_i \in D$, then we have

$$M_j - m_j \leq M - m = \sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \quad (43)$$

$$\Delta \alpha_j = \alpha(x_j) - \alpha(x_{j-1}) \leq |\alpha(x_j) - \alpha(y_i)| + |\alpha(y_i) - \alpha(x_{j-1})| \quad (44)$$

$$< 2c_k \varepsilon \quad (45)$$

Therefore,

$$U(P_\varepsilon, f, \alpha) - L(P_\varepsilon, f, \alpha) = \sum_{j=1}^{m_\varepsilon} (M_j - m_j) \Delta \alpha_j \quad (46)$$

$$= \sum_{j=1, y_{i_j} \in D}^{m_\varepsilon} (M_j - m_j) \Delta \alpha_j + \sum_{j=1, y_{i_j} \notin D}^{m_\varepsilon} (M_j - m_j) \Delta \alpha_j \quad (47)$$

$$\leq 2c\varepsilon \underbrace{\sum_{j=1, y_{i_j} \notin D}^{m_\varepsilon} \Delta \alpha_j}_{ss} + (M - m) \sum_{j=1, y_{i_j} \in D}^{m_\varepsilon} \Delta \alpha_j \quad (48)$$

$$(49)$$